

1MT495 Current Issues in Applied International Economics: Intertemporal Approach

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Outline

- 1 Reminder: Balance of Payments (BoP)
- 2 Motivation and Core Concepts
- 3 Basic Models
- 4 Extensions: Investment and Global Models
- 5 Applications and Limitations
- 6 Criticisms of the Intertemporal Approach
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Disclaimer

- My goal is to give you a broad overview of the **key models and ways of thinking** in modern international economics.
- **Focus on intuition and economic logic:** While the models use mathematics, we'll emphasize the main ideas, not detailed proofs.
- I will focus on the **main ideas and ways of thinking** in modern international economics.
- Most of the material is not my own I have collected and adapted lecture notes from people much smarter than me.
- **Source Check:** At the end of each lecture, I'll share the sources. Please check the original materials if you'd like to explore the topic in more depth.

Key Definitions and Concepts: Balance of Payments (BoP)

Definition of BoP

A systematic accounting statement capturing all economic transactions between residents of the domestic country and residents of the rest of the world that occurred over a given period.

- Most commonly annual frequency lower frequency creates issues with seasonality.
- **A flow statistic, referring to a specific period**
- The key document for compiling the BoP is the IMF publication *Balance of Payments and International Investment Position* (6th edition) BPM6.
- The main change from BPM5 to BPM6 includes, among others, an increased emphasis on the investment position, which appears in a different recording of capital flows in the financial account.

Overall Structure of the BoP (according to BPM6)

The defined structure (BPM6) is:

- 1 Current Account (CA)
- 2 Capital Account (KA)
- 3 Net Errors and Omissions (NEO)
- 4 Financial Account (FA)

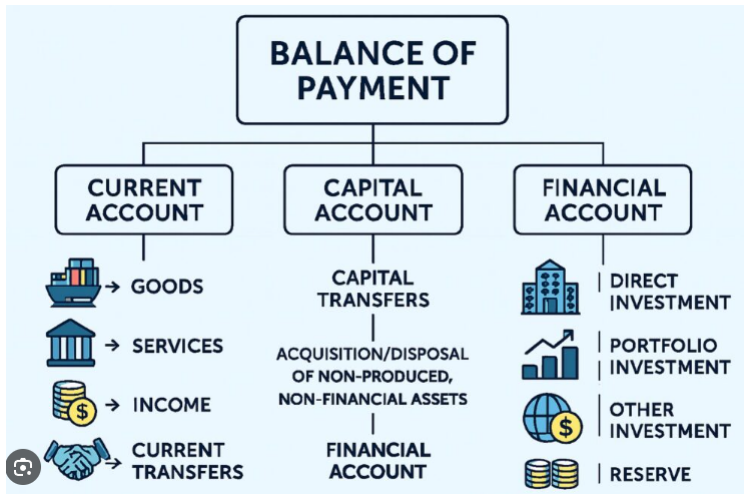
The BoP Identity Components:

- **Above-the-line items (Net Lending/Borrowing):** $CA + KA + NEO$
- **Below-the-line items (Financing):** Financial Account (FA) $\pm$ Changes in Reserve Assets

The Fundamental BoP Identity

$$CA + KA + NEO = FA_{\text{Net Change}}$$

BoP Structure (BPM5)



BoP Structure (BPM6)

Current account	Credit [+]	Debit [-]
Goods	Export	Import
Services	Export	Import
Primary income	Import (Income, inflow)	Export (Expenditures, outflow)
Secondary income	Import (Income, inflow)	Export (Expenditures, outflow)

Capital account	Credit [+] Import (Income, inflow)	Debit [-] Export (Expenditures, outflow)
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Financial account	Foreign ASSETS movements	Foreign LIABILITIES movements
Direct investments	Rise [+], decline [-]	Rise [+], decline [-]
Portfolio investments	Rise [+], decline [-]	Rise [+], decline [-]
Financial derivatives	Rise [+], decline [-]	Rise [+], decline [-]
Other investments	Rise [+], decline [-]	Rise [+], decline [-]
Reserve assets	Rise [+], decline [-]	-----

Errors and omissions	Credit [+]	Debit [-]
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Differences Between BPM5 and BPM6

Only for cases where you read older papers/literature.

BoPM5

Current account (*běžný účet*)
 Goods and services (*výkon. balance*)
 Goods (*obchodní balance; zboží*)
 Services (*balance služeb; služby*)
 Income (*balance výnosů*)
 Current transfers (*balance transferů*)
Capital and financial account
 Capital account (*kapitálový účet*)
 Financial account (*finanční účet*)
 Direct investment (*př. zahr. inv.*)
 Portfolio investment (*port. inv.*)
 Financial derivatives (*fin. deriv.*)
 Other investment (*ostat. inv.*)
Reserve assets (*„devizové rezervy“*)
Errors and omissions (*chyby a opomenutí*)

BoPM6

Current account (*běžný účet*)
 Goods and services (*výkon. balance*)
 Goods (*obchodní balance; zboží*)
 Services (*balance služeb; služby*)
 Primary income (*prvotní důchody*)
 Secondary income (*druhotné důchody*)
Capital account (*kapitálový účet*)
Financial account (*finanční účet*)
 Direct investment (*přímé zahraniční investice*)
 Portfolio investment (*portfoliové investice*)
 Financial derivatives (other than reserves) (*finanční deriváty*)
 Other investment (*ostatní investice*)
 Reserve assets (*rezervní aktiva*)
Net errors and omissions (*čisté chyby a opomenutí*)

More on the Financial Account of the Balance of Payments

- **Foreign Direct Investment (FDI)** investment representing a stake of **10%** or more in equity capital, plus long-term debt between affiliates. (Less volatile)
- **Portfolio Investment** equity or debt securities that do not meet the 10% threshold. Highly volatile.
- **Other Investment** long-term loans (firms, banks, government) and short-term capital flows (deposits, T-bills, commercial paper). Most volatile, essential for liquidity.
- **Financial Derivatives**
- **Reserve Assets** central bank operations with the rest of the world (e.g., currency intervention, gold holdings, SDRs). **Recorded as a flow** (change in stock) in the FA.

Errors and Omissions

- Errors and omissions arise due to, for example, short-term capital movements, (il)legal movement of persons across borders, valuation errors, etc.
- If this item is positive (negative), one may infer that the values on a certain side of the current and financial account (i.e., credit and debit, or changes in assets and liabilities) are underestimated (overestimated).

Motivation: The Intertemporal Approach

- The **intertemporal approach to international trade** views the current account as the outcome of **forward-looking saving and investment decisions**.
- It explains **how countries trade goods and assets over time**, responding to income shocks and world interest rates.
- This approach is **dynamic**, allowing for persistent Current Account surpluses or deficits, contrasting sharply with static trade models (e.g., Heckscher-Ohlin).

What is the Intertemporal Approach?

The intertemporal approach views international trade as occurring **through time**, focusing on borrowing, lending, saving, and investment between countries.

- Unlike static models (e.g., **Ricardian** or **Heckscher-Ohlin**), it explains **persistent trade imbalances** (deficits or surpluses).
- Countries use international capital markets to **smooth consumption** in response to temporary shocks.
- **Key Identity:** $CA_t = S_t - I_t = \Delta NFA_t$
- Current Account (CA) = Domestic Saving (S) - Domestic Investment (I) = Change in Net Foreign Assets (ΔB_t).

Key Definitions and Concepts

- **Current Account (CA) Definition:** $CA_t = TB_t + rB_{t-1}$ where $TB_t = Y_t - C_t - I_t - G_t$ (Trade Balance), and rB_{t-1} is **Net Investment Income** on previously accumulated NFA = B_{t-1} .
- **Intertemporal Budget Constraint (IBC):** $\sum_{t=0}^{\infty} \frac{CA_t}{(1+r)^t} = 0$
This states that the **present value of all future Current Accounts must sum to zero** for a country to be solvent.
- **Temporary vs. Permanent Shocks:** Temporary shocks lead to large CA swings for smoothing; permanent ones adjust consumption with minimal CA impact.
- **Autarky vs. Open Economy:** In autarky, $C_t \leq Y_t$; with openness, agents borrow/lend at world rate r^* .
- **Consumption Smoothing:** Households prefer stable consumption paths, borrowing during low-income periods and saving during high ones.

Two-Period Small Open Economy Model: Assumptions

Assumptions for the small open endowment economy:

- Two periods (1 and 2), single good
- Exogenous endowments Y_1, Y_2
- Small open economy: Exogenous world interest rate $r^* = r$
- Utility: $U = u(C_1) + \beta u(C_2)$, with $u' > 0, u'' < 0$.
- Initial assets $B_0 = 0$, terminal $B_2 = 0$ (transversality)
- Perfect foresight, no government or investment

Two-Period Small Open Economy Model: Budget Constraints

Sequential budget constraints:

- Period 1: $C_1 + B_1 = (1 + r_0)B_0 + Y_1$ (simplifies to $C_1 + B_1 = Y_1$ if $B_0 = 0$).
- Period 2: $C_2 = (1 + r)B_1 + Y_2$.

Intertemporal budget constraint (IBC):

$$C_1 + \frac{C_2}{1 + r} = Y_1 + \frac{Y_2}{1 + r}$$

This equates the present value of consumption to the present value of endowments.

Two-Period Small Open Economy Model: Utility Maximization

Maximize $U = u(C_1) + \beta u(C_2)$ subject to IBC.

Lagrangian:

$$\mathcal{L} = u(C_1) + \beta u(C_2) + \lambda \left(Y_1 + \frac{Y_2}{1+r} - C_1 - \frac{C_2}{1+r} \right)$$

First-order conditions (FOCs):

$$\frac{\partial \mathcal{L}}{\partial C_1} = u'(C_1) - \lambda = 0 \quad \implies u'(C_1) = \lambda$$

$$\frac{\partial \mathcal{L}}{\partial C_2} = \beta u'(C_2) - \frac{\lambda}{1+r} = 0 \quad \implies \beta u'(C_2) = \frac{\lambda}{1+r}$$

Two-Period Small Open Economy Model: Euler Equation

Combining FOCs yields the **Euler equation**:

$$\frac{u'(C_1)}{u'(C_2)} = \beta(1 + r)$$

- This equation equates the marginal rate of substitution (MRS) between consumption today and consumption tomorrow with the effective return on saving.
- **Optimal Consumption Smoothing:** Consumption is chosen such that the benefit of saving (discounted return $\beta(1 + r)$) equals the cost (foregone utility $u'(C_1)/u'(C_2)$).

Two-Period Small Open Economy Model: Solution

Example (Log Utility)

Assume $u(C) = \ln C$.

- **Case for Perfect Smoothing:** If the subjective discount factor equals the market discount factor ($\beta = 1/(1+r)$), then $C_1 = C_2$.

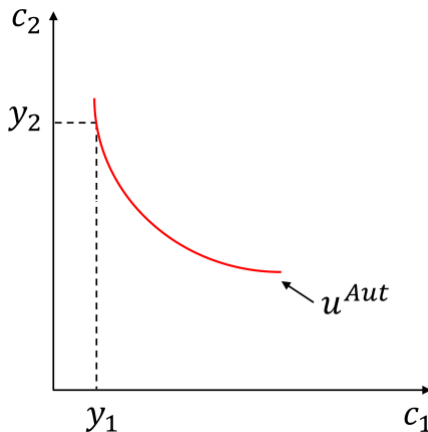
$$C_1 = C_2 = \frac{(1+r)Y_1 + Y_2}{2+r}$$

- **General Log Case** (where $\beta(1+r) \neq 1$):

$$C_1 = \frac{1}{1 + \beta(1+r)} \left(Y_1 + \frac{Y_2}{1+r} \right), \quad C_2 = \beta(1+r)C_1$$

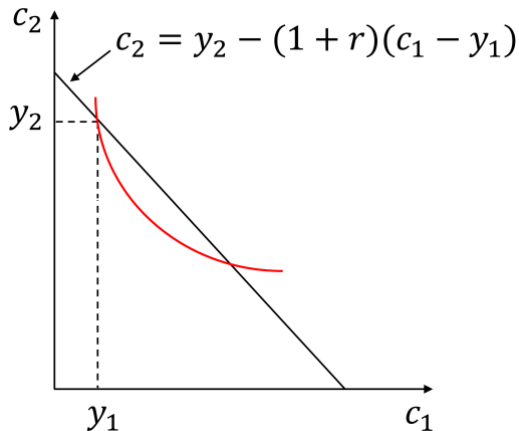
Graphical Representation: Autarky

Consume endowments and receive some autarky level of utility



source: <https://danwales.com/wp-content/uploads/2018/10/lecture12.pdf>

Graphical Representation: Introduction of Trade



source: <https://danwales.com/wp-content/uploads/2018/10/lecture12.pdf>

Autarky vs. Open Economy

- **Autarky:** Consumption equals endowment ($C_t = Y_t$). The country is limited to the local opportunity set (MRS).
- **Open Economy:** Access to the world interest rate r allows the country to shift its budget line (IBC) to a better opportunity set.
- **The Choice:** The country moves to the highest indifference curve tangent to the new budget line (PV of endowments).
- If $r^{aut} > r$, the country borrows (runs a CA deficit in $t = 1$) to bring consumption forward.

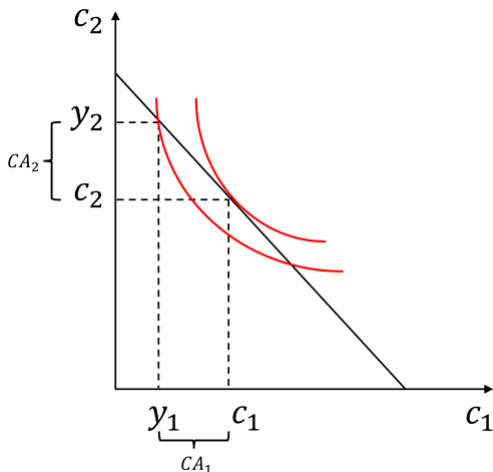
Optimal Consumption and the Current Account

- The optimal consumption bundle (C_1, C_2) is found using the IBC and Euler equation.
- The Current Account in period 1 is the difference between endowment and optimal consumption:

$$CA_1 = Y_1 - C_1 = B_1$$

- **Consumption Smoothing:** If Y_1 is temporarily higher than Y_2 , C_1 is less than Y_1 , leading to $B_1 > 0$ (CA surplus) to finance higher C_2 .

Graphical Representation: New Equilibrium and Current Accounts



source: <https://danwales.com/wp-content/uploads/2018/10/lecture12.pdf>

Trade Balance vs. Current Account

- **Trade Balance (TB):** Trade in goods and services only.

$$TB_t = Y_t - C_t - I_t - G_t$$

- **Current Account (CA):** Trade in goods/services PLUS Net Investment Income.

$$CA_t = TB_t + rB_{t-1}$$

- **Implication:** A country can run a persistent **Trade Deficit** ($TB < 0$) in the long run if it holds sufficient positive net foreign assets ($B_{t-1} > 0$), as the investment income covers the deficit.

Responses to Income Shocks: The Current Account as a Smoother

Table: Impact of Different Income Shocks on CA

Shock Type	CA Impact	Intuition
Temporary $\uparrow$ (e.g., $\Delta Y_1 > 0$)	Surplus ($\uparrow B_1$)	Save the windfall to smooth C
Permanent $\uparrow$ ($\Delta Y_1 = \Delta Y_2 > 0$)	Minimal/No Change	C adjusts fully; minimal need to save/borrow
Anticipated Future $\uparrow$ ($\Delta Y_2 > 0$)	Deficit ($\downarrow B_1$)	Borrow today to smooth C up

Two-Period Small Open Economy Model: Summary

In a small open economy (exogenous $r = r^*$):

- The **IBC** determines the optimal present value of consumption, $PV(C)$.
- The **Euler Equation** $u'(C_1) = \beta(1 + r)u'(C_2)$ determines the optimal distribution of consumption over time, C_1 and C_2 .
- **Current Account (CA):** $CA_1 = Y_1 - C_1 = B_1$ is determined **endogenously** to facilitate consumption smoothing.
- **Example:** If current endowment Y_1 is temporarily higher than Y_2 , the country saves (runs a CA surplus, $B_1 > 0$) to smooth consumption.
- **Global Model Implication:** For a large economy, shocks propagate globally via the endogenous adjustment of the world interest rate r .

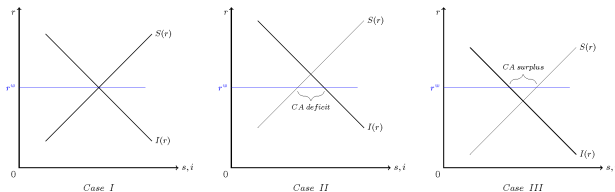
Extending the Model: Investment and the CA

When domestic investment I_t is introduced, it also depends on the world interest rate r .

- The Current Account becomes the difference between **Domestic Saving** and **Domestic Investment**:

$$CA_t = S_t(r) - I_t(r)$$

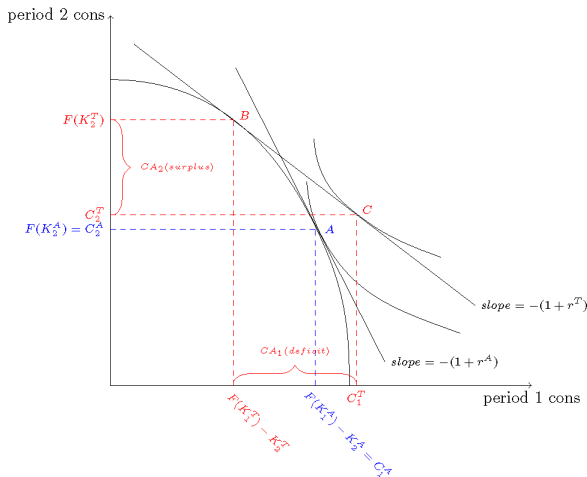
- **CA Determinants:** A country runs a CA deficit if I opportunities are high relative to S (e.g., high productivity or temporary low r).



source:

https://people.brandeis.edu/~ghall/econ303/build/html/open_saving_invest.html

Investment and the CA: New Equilibrium and Current Accounts



source:

https://people.brandeis.edu/~ghall/econ303/build/html/open_saving_invest.htm

Extending the Model: Infinite Horizon Models

Extends the model to infinite periods ($t = 0$ to ∞).

- **Utility:** $\sum_{s=t}^{\infty} \beta^{s-t} U(C_s)$
- **Stochastic Euler Equation:** Incorporates uncertainty E_t :

$$U'(C_t) = \beta(1+r)E_t U'(C_{t+1})$$

- **Intertemporal Budget Constraint (IBC):** The fundamental constraint that ensures national solvency (No-Ponzi condition):

$$\sum_{t=0}^{\infty} \frac{CA_t}{(1+r)^t} = 0$$

- Persistent deficits must eventually be offset by future surpluses to maintain solvency (No-Ponzi condition).
- In long-run **steady-state** equilibrium, the change in net foreign assets is zero ($\overline{CA} = 0$).
- However, $\overline{TB} = -\bar{r} \bar{B}$ implies a country can run a **long-run trade deficit** ($\overline{TB} < 0$) if it has a sufficiently large, positive net foreign asset position ($\bar{B} > 0$).

Global Equilibrium: The Market for International Capital

For models with large economies, the world interest rate r is endogenous.

- **Global Constraint:** In global equilibrium, the sum of all country Current Accounts must be zero:

$$CA_t + CA_t^* = 0$$

- **Determination of r :** The world interest rate r^* adjusts until global supply of savings equals global demand for capital (investment + borrowing).
- **Shocks:** Shocks in one large country (e.g., a fiscal stimulus increasing borrowing) propagate globally by changing r^* .

World Economy Model (Two Countries)

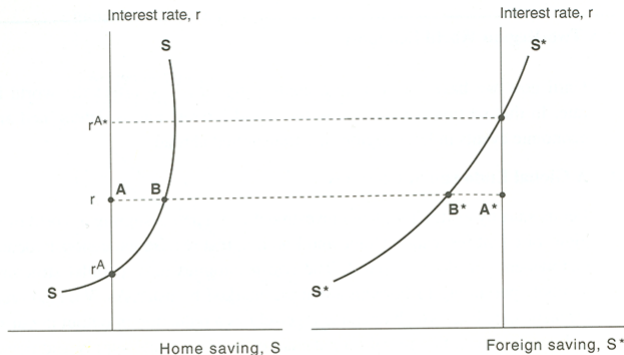
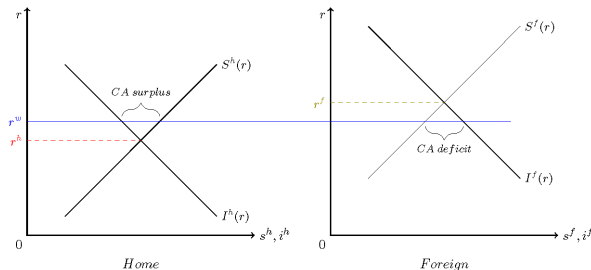


Figure 1.5
Global exchange equilibrium

source: Ida Wolden Bache lecture notes

World Economy Model (The Determination of the World Interest Rate)

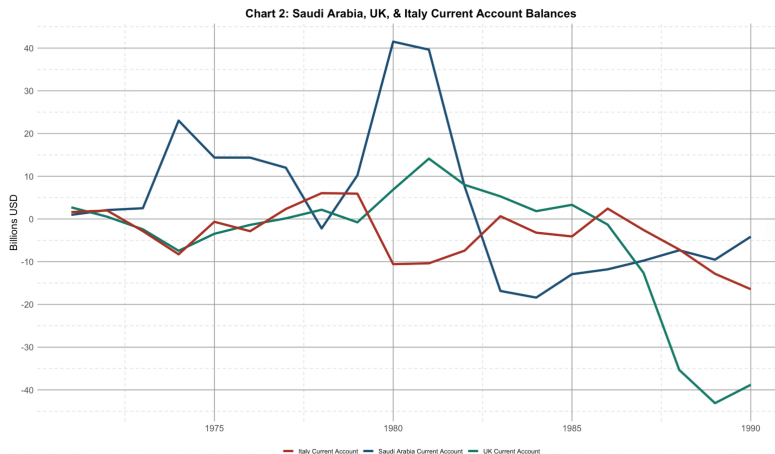


source: <https://people.brandeis.edu/~ghall/econ303>

Applications: Oil Shocks, Immigration, Global Imbalances and Debt Crises

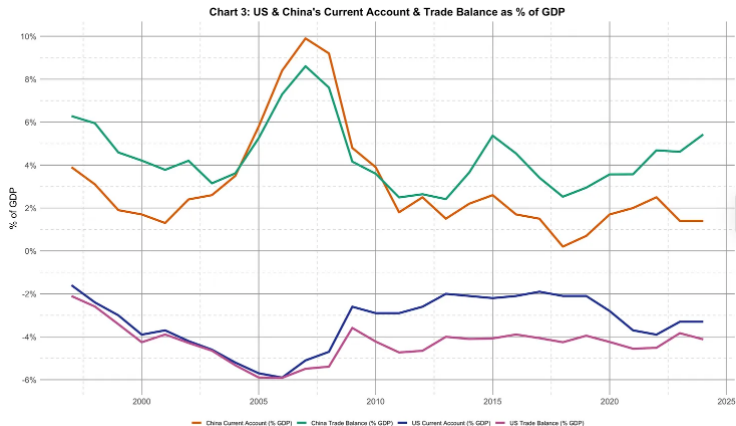
- 1970s Oil Crises: Importers ran deficits by borrowing to smooth consumption; exporters surpluses by lending.
- Israel's 1990s Immigration: Temporary income drop led to CA surplus for smoothing.
- Norway's Oil Wealth: High future productivity prompted surpluses; fiscal rule spends annuity value.
- U.S.-China Imbalances: U.S. deficits from low saving; China surpluses from high growth expectations.
- Sovereign Debt: Unsustainable if required surpluses exceed feasible NX; e.g., Greece needed surplus but ran deficits.

An Example: Oil Price Shocks of the 1970s



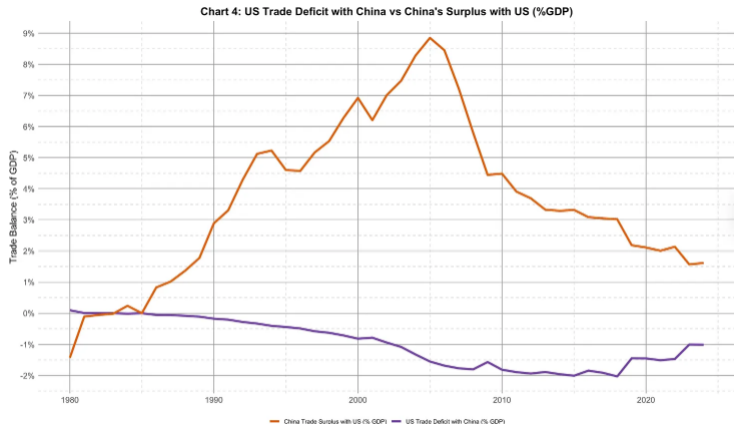
source: <https://gianlucabenigno.substack.com/p/background-note-on-trade-and-trade>

The Current Imbalances 1 (US and China)



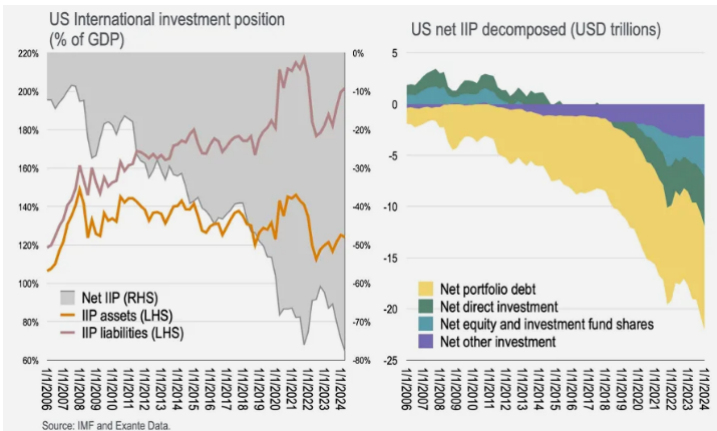
source: <https://gianlucabenigno.substack.com/p/background-note-on-trade-and-trade>

The Current Imbalances 2 (US and China)



source: <https://gianlucabenigno.substack.com/p/background-note-on-trade-and-trade>

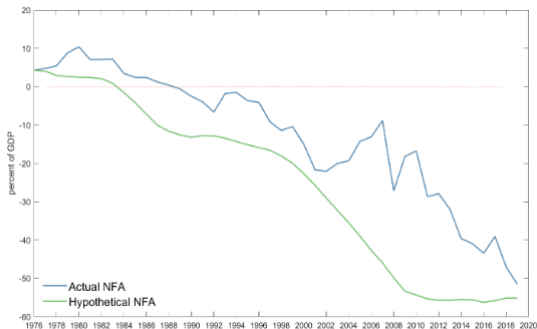
The Current Imbalances 3 (US International Investment Position - Exorbitant Privilege?)



source:

<https://moneyinsideout.substack.com/p/united-states-changing-net-iip>

The Current Imbalances 3 (The US NIIP and the Hypothetical NIIP Without Valuation Changes Since 1976)

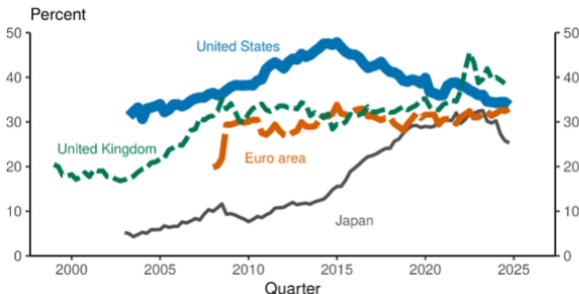


source:

https://people.brandeis.edu/~ghall/econ303/build/html/niip_paradox.html

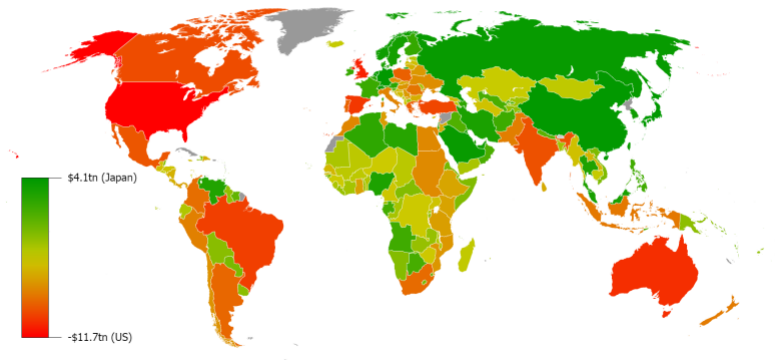
The Current Imbalances 4 (Securities - Foreign Share)

4b. Share of general government debt securities held by foreign investors



source: <https://www.federalreserve.gov/econres/notes/feds-notes/the-international-role-of-the-u-s-dollar-2025-edition-20250718.html>

The Current Imbalances 5 (World)



source: Dan Wales Lecture notes

Introduction to Criticisms: A "Real" Model in a "Financial" World

The intertemporal approach (IA) provides a valuable "real" framework (goods, saving/investment) but struggles to capture modern financial dynamics. Key issues stem from its reliance on national accounting aggregates:

- **Oversight 1: Flows and Volatility.** Failure to distinguish between large **gross** capital flows and small **net** flows.
- **Oversight 2: Concepts.** Conflation of **saving** (ex post, non-financial) with **financing** (ex ante, monetary/credit).
- **Oversight 3: Mechanism.** Ignorance of **endogenous money creation** (loans create deposits) in modern banking.

These critiques (Borio and Disyatat, 2011; Avdjiev et al., 2015; Felipe, 2024) challenge the model's relevance for understanding financial crises and global imbalances.

Criticism 1: Net vs. Gross Capital Flows

The Intertemporal Approach (IA) focuses exclusively on **net capital flows** (the Current Account), capturing only net resource transfers. This overlooks the massive **gross flows** crucial for financial stability.

- **IA Focus: Net Flows (Δ NFA)**

- Represents the net resource transfer over time.
- Reveals little about financing patterns; excludes offsetting asset/liability transactions.
- Poor indicator of **financial stress**.

- **Financial Reality: Gross Flows**

- Gross flows are **orders of magnitude larger** (e.g., 435% of US GDP in 2010).
- They reveal bilateral exposures and **global intermediation patterns**.
- Gross flows are highly **volatile**; their collapse (e.g., 75% drop in US inflows during the GFC) is the true indicator of crisis.

Conclusion: Emphasizing net flows provides a misleading picture of financial stability and obscures the gross liabilities that generate systemic risk (Borio and Disyatat, 2011). Empirical tests based on net data, like Feldstein-Horioka, suffer from a "pseudo bias" (Feldstein, 2004).

Criticism 2: Saving vs. Financing Conflation (Concept)

The IA wrongly equates **saving** (a non-financial, ex post residual) with **financing** (an ex ante, cash-flow concept involving credit).

- **Saving:** $S = Y - C - G$. It is an **ex post** national accounts residual, tied to the goods market.
- **Financing:** Access to purchasing power through **credit or money creation**. It is an **ex ante** money market concept.
- **The Conflation:** The IA treats saving as the necessary source of funds for investment/deficits.

Reality Check: Credit booms (the source of financing) can occur equally well with Current Account surpluses (Japan, 1980s) or deficits (US, pre-GFC), proving that the two concepts are logically and empirically **decoupled**.

Criticism 2: Saving vs. Financing Conflation (Implications)

This conceptual conflation leads to critical misinterpretations and policy errors:

- **Misinterpretation:** Deficits are attributed to a foreign "global saving glut" (due to foreign saving being channeled) rather than **domestic credit creation** (due to domestic financing).
- **Policy Error:** Over-focusing on current account imbalances (the saving side) while ignoring mounting **financial vulnerabilities** from gross exposures, maturity mismatches, and domestic credit creation.
- **Core Flaw:** The model overlooks the ability of the financial system to generate purchasing power independently of underlying real flows or saving behavior.

Criticism 3: Endogenous Money and the Reversal of Causality

A core flaw is the IA's adherence to the **Loanable Funds** view (saving $\implies$ investment), ignoring that modern banking operates via **endogenous money creation** (Borio & Disyatat, 2011).

- **IA Assumption:** Saving precedes and funds investment/lending.
- **Banking Reality:** Banks create purchasing power "out of thin air." A loan is made, and the deposit (new money) is created simultaneously on the balance sheet.

Bank Loan $\implies$ New Deposit $\implies$ Financing

- **Causal Reversal:** Investment/spending creates the financing (deposits) ex ante, and this spending then **generates saving ex post** (as a residual in the national accounts).

This mechanism completely decouples the economy's ability to finance deficits from the availability of prior savings.

Criticism 3: Policy Impact of Endogenous Money

Ignoring endogenous money leads to flawed conclusions about international financing:

- **Deficit Financing:** Current account deficits are often financed by **domestic banks** creating deposits (e.g., US dollar credit creation), not by foreign surplus countries transferring real resources.
- **Focus Shift:** The IA's "real" approach veils monetary factors, leading to a focus on saving imbalances when the actual source of instability is the financial elasticity (the speed/scale of credit creation).
- **Crisis Lesson:** Financial crises stem from **credit booms** (financial factors), not necessarily from "saving gluts" or current account imbalances (real factors).

Overall Implications and Conclusions

- **Model Failure:** The IA's exclusive reliance on **net flows** and a **real analysis** is insufficient in a monetary economy characterized by large, volatile gross financial dynamics.
- **Policy Misguidance:** This framework generates futile empirical results (e.g., the "pseudo bias" in Feldstein-Horioka) and diverts policy focus toward stabilizing net imbalances rather than tackling underlying financial vulnerabilities (credit, maturity mismatches).
- **Recommendation: Shift the Paradigm.** For a better understanding of global financial vulnerabilities and crises, economists should:
 - ① Shift analytical focus from **Net to Gross** financial flows.
 - ② Clearly distinguish between **Saving** (ex post) and **Financing** (ex ante).
 - ③ Incorporate **banking sector models** featuring endogenous money creation.

Summary: The Intertemporal Approach (IA) in Perspective

The IA provides a rigorous, dynamic, and efficient framework for understanding global resource allocation over time, but its reliance on **real flows** limits its applicability in modern **financial** markets.

- **Core Insight (The Good):** The IA successfully models current account imbalances as rational, forward-looking responses to optimize consumption smoothing and investment opportunities.
- **Policy Guidance:** It teaches that deficits are not inherently "bad" if they finance productive investment or smooth consumption after a temporary shock.
- **Theoretical Foundation:** The framework is based on clear micro-foundations (IBC and Euler Equation) under the assumption of perfect capital markets and rational agents.
- **Limitations (The Critique):** Its reliance on net flows and the saving $\implies$ financing identity fails to capture financial fragility driven by gross capital flows and endogenous credit creation.

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- Gianluca Benigno lecture notes (Background Note on Trade and Trade Imbalances)
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- Obstfeld and Rogoff (The Intertemporal Approach to the Current Account)
- Dan Wales Lecture notes <https://danwales.com/teaching/>
- Ida Wolden Bache Lecture notes
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Thank You

Questions?